

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Are binary and decimal number systems really essential in programming?



Learning Outcome:

Exchange information about logic, its principles, and methods in simple spoken and written situations.
Describe the development and software programming in simple spoken and written situations.

Ask your partner:

- Why do you think the decimal and binary number systems are important in programming?
- In which contexts do you need to use them? Why?

Can Do Statements:

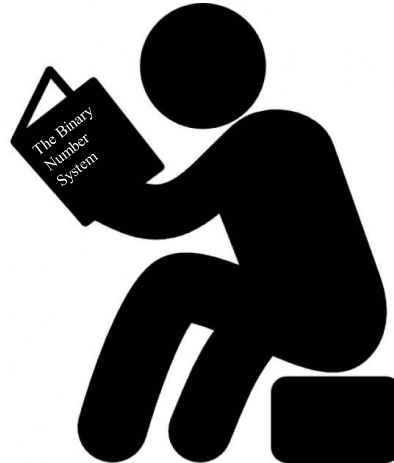
- I can say how to understand the decimal number system.
- I can say how to understand the binary number system.
- I can answer some questions about the binary number system in programming.





Let's Explore!

Do you remember everything about binary and decimal number systems?



Instructions: Read the text below.

Teacher: Today, we continue with the binary number system. Can you tell me how to understand it? What about the decimal number system? Do you know any example?

Paula: Well, I remember some basic concepts I learned at school. I think that to understand the binary system we can start with the decimal one, which works with units of ten, or "Base 10", as we know it. To represent numbers, this system needs the digits from 0 to 9. I remember dividing a number into three columns: column H for hundreds, T for tens and O for ones, so that, in that order, the number 184 is 1-hundreds plus 8-tens, plus 4-ones.

Teacher: Good! How can you depict that number in the corresponding equation?

Paula: 184 is $\{(1 \cdot 10^2) + (8 \cdot 10^1) + (4 \cdot 10^0)\}$, taking into account that the ones column is 10^0 , the tens column, 10^1 and the hundreds column, 10^2 .

Teacher: Correct! Luis, what can you tell me about the binary number system?

Luis: It differs from the decimal system in some ways. First, we know it as "Base 2" because it only works with 2 digits. Each digit has two possible values: 0 or 1.

Paula: Is that what we call a "bit"?

Luis: Yes, "bit" is the short form of "Binary Digit". Each bit has to be multiple of 2 if we talk about a binary number system.

Teacher: Great! But, why do you think these topics are important in programming?

Paula: Well, you use the binary system to express data for computer programming! With this system you can execute a lot of operations in computing as solving software engineering problems, extracting information from bit fields and reading raw data.



**Let's Practice!**

Let's clarify some concepts!

Instructions: Select the correct answer to complete each sentence.

1. We can start with the decimal system, it works with units of ten is called _____.
 - a. "Base 10"
 - b. "Base 1"
 - c. "Base 2"
2. To represent numbers, this system needs the digits _____.
 - a. by 0 to 9.
 - b. from 0 to 9.
 - c. To 0 from 9.
3. _____ a number in three columns?
 - a. Can I dividing
 - b. Can we divide
 - c. Can they divided
4. _____ tell me about the binary and decimal number systems?
 - a. Can you what
 - b. What can you
 - c. What you can
5. _____ from the decimal system in some ways.
 - a. Binary can differs
 - b. Binary can differ
 - c. Binary can differed

**Let's Practice!**

Can you tell me what the last class was about?

Instructions: Read the questions. Underline all the mistakes in the questions.

1. Can you mentioned an example of a decimal number system? Yes! The number 165 is 1-hundreds plus 6-tens, plus 5-ones.
2. Must you gave me an example of a binary number system? Yes! The number 10100 represents one "group" of 16, 0 groups of 8, 1 group of 4, 0 groups of 2 and the value 0.
3. Can you telling my about the binary number system of "Binary Digit"? Yes, it is bit.





Extension Activity

Decimal and Binary Digits

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about the main uses of representations of decimal and binary digits.
- Write down a paragraph in which you mention important aspects of them.
- Share the paragraph with your instructor

Self-Assessment

Yes

No

1. I look for the main uses of representations of decimal and binary digits.

☐ ☐

2. I write down a paragraph important aspects.

☐ ☐

3. I share the paragraph with the instructor.

☐ ☐


Target Words

- **raw data** (noun): primary data that comes from one source
e.g. You can execute a lot of operations in computing as solving software engineering problems, extracting information from bit fields and reading raw data.
- **digit** (noun): a number
e.g. Decimal system needs the digits from 0 to 9.
- **equation** (noun): a mathematic symbolic representation.
e.g. How can you depict that number in the corresponding equation?
- **differ from** (verb): to be different from something
e.g. Binary system differs from the decimal system in some ways.
- **depict** (verb): to represent
e.g. How can you depict that number in the corresponding equation?
- **extract** (verb): to remove, to collect
e.g. you can execute a lot of operations in computing as solving software engineering problems, extracting information from bit fields and reading raw data.



Learning Points

Verb Can, and Simple Present Tense to Ask Questions on Programming

You formulate questions with **can** and the **simple present tense** to get information about present facts; in this case, to collect information about programming.

Use the modal verb **can** to express permission, possibility or ability. To formulate questions, have in mind the following structure:

Can + subject+ verb + compliment?

Can you use Boolean expressions in this context?

Use the simple present to ask questions about present facts. Don't forget the auxiliary verb **Do / Does** when formulating the question:

Do/Does+ subject + verb+ compliment?

Do you know how to store a number 1 through 100?

Does she understand how the decimal system works?

Examples

- *How can you represent that number in the corresponding equation?*
- *Do you know any example?*
- *Why do you think these topics are important in programming?*



Learning Points

Representation of Decimal and Binary Digits

Binary and decimal systems are widely used in computing science. Some aspects of this one requires the knowledge of these systems, which is why it is crucial to understand how they work.

In the following chart, you'll see the representation of these two number systems:

Decimal Digits	Binary Digits	Representation
0	0	Value of 0
1	1	Value of 1
2	10	One "group" of 2 followed by the value 0
3	11	One "group" of 2 followed by the value 1
4	100	One "group" of 4 followed by 0 groups of 2 followed by the value 0
5	101	One "group" of 4 followed by 0 groups of 2 followed by the value 1
6	110	One "group" of 4 followed by 1 group of 2 followed by the value 0
7	111	One "group" of 4 followed by 1 group of 2 followed by the value 1
8	1000	One "group" of 8, 0 groups of 4, 0 groups of 2 and the value 0
...	...	...
20	10100	One "group" of 16, 0 groups of 8, 1 group of 4, 0 groups of 2 and the value 0
...	...	...
100	1100100	1 group of 64, 1 group of 32, 0 groups of 16, 0 groups of 8, 1 group of 4, 0 groups of 2 and 0
...	...	...
125	1111101	1 group of 64, 1 group of 32, 1 group of 16, 1 group of 8, 1 group of 4, 0 groups of 2 and 1